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21ADE25 BUSINESS ANALYTICS

ASSIGNMENT -3
(https://github.com/prithvi47/Business_Analysis_Assign3.git)

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GitHub Repo --
https://github.com/prithvi47/Business_Analysis_Assign3.git

Airline Passenger Traffic Forecasting

A Time Series Analysis for Operational Planning and Demand Management

Abstract

This report presents a comprehensive forecasting framework for airline passenger demand using the classic AirPassengers dataset (1949–1960). The primary objective is to predict future passenger volumes to support strategic decisions in fleet allocation, crew scheduling, pricing, and maintenance planning. After confirming the presence of a strong upward trend and multiplicative seasonality, we apply the Holt-Winters triple exponential smoothing model. The model achieves a Mean Absolute Percentage Error (MAPE) of 6.8% on a 12-month test period, indicating excellent predictive performance. Seasonal decomposition reveals that passenger demand in July and August is 28% above the annual average, while February and November are 18% below average. Operational recommendations include increasing flight capacity by 25-30% during summer peaks, scheduling heavy maintenance in low-demand months, and implementing dynamic pricing. The report concludes by suggesting hybrid approaches that combine Holt-Winters with SARIMA, Prophet, or machine learning residual correction to further improve accuracy.

1. Introduction

Air passenger traffic is inherently volatile, influenced by economic cycles, seasonal holidays, fuel prices, and unforeseen events (e.g., pandemics). Accurate forecasting enables airlines to optimise resource utilisation, minimise operational costs, and enhance customer satisfaction. This project uses the monthly international airline passenger counts from 1949 to 1960 – a benchmark time series in forecasting literature – to demonstrate a robust methodology that can be adapted to real-world airline data.

The specific goals are:

- Analyse the underlying trend and seasonal patterns.
- Apply the Holt-Winters (triple exponential smoothing) model, which explicitly handles trend and seasonality.

- Quantify forecast accuracy using standard error metrics.
- Translate statistical outputs into actionable operational plans.
- Propose advanced hybrid models for future improvements.

2. Data Description

Source: The AirPassengers dataset is built into the statsmodels library and originally published by Box, Jenkins & Reinsel (1994). It records the monthly total number of international airline passengers (in thousands) from January 1949 to December 1960.

Feature	Value
Number of observations	144 months (12 years)
Time range	1949-01-31 to 1960-12-31
Minimum passengers	104 thousand (April 1950)
Maximum passengers	622 thousand (July 1960)
Mean monthly passengers	280 thousand
Total passengers over period	40.3 million

The data exhibit a clear upward trend and a seasonal pattern where summer months (July–August) consistently show higher traffic, while winter months (February, November) show lower traffic.

3. Exploratory Analysis and Stationarity Testing

3.1 Visual Inspection

The time series plot (Figure 1) reveals:

- **Trend**: Steady linear growth from about 120 thousand in 1949 to over 600 thousand in 1960.
- **Seasonality**: The amplitude of seasonal swings grows with the trend – indicative of **multiplicative seasonality**.
- **Noise**: Irregular fluctuations around the seasonal pattern, but no obvious outliers.

3.2 Seasonal Decomposition

Using a multiplicative decomposition (period = 12 months), we isolated:

- **Trend component**: Smoothly increasing, confirming long-term market growth.
- **Seasonal component**: A repeating 12-month pattern with a peak in July (factor ≈ 1.28) and a trough in November (factor ≈ 0.82).
- **Residual (irregular)**: Random noise with constant variance, suggesting that the multiplicative model adequately captures the dynamics.

3.3 Augmented Dickey-Fuller (ADF) Test

The ADF test checks for unit root (non-stationarity).

Statistic	Value
ADF test statistic	0.82
p-value	0.99
Critical value (5%)	-2.88

Conclusion: The p-value is far above 0.05, so we cannot reject the null hypothesis of non-stationarity. This confirms the presence of trend and seasonality, justifying the use of a seasonal model like Holt-Winters.

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4. Model Development: Holt-Winters (Triple Exponential Smoothing)

4.1 Why Holt-Winters?

- **Additive trend** – captures the steady linear increase.
- **Multiplicative seasonality** – reflects that seasonal fluctuations become larger as traffic grows.
- **Simple implementation** – requires only historical data, no exogenous variables.
- **Interpretable parameters** – smoothing constants α, β, γ provide insight into how the model adapts to recent changes.

4.2 Train-Test Split

- **Training set**: January 1949 – December 1958 (120 months)
- **Test set**: January 1959 – December 1960 (24 months)

We deliberately kept two full years for out-of-sample evaluation.

4.3 Model Calibration

The model was fitted using maximum likelihood estimation. The optimised smoothing parameters are:

Parameter	Symbol	Value	Interpretation
Level smoothing	α	0.42	Moderate responsiveness to recent level changes
Trend smoothing	β	0.009	Very slow adaptation of trend (almost constant)
Seasonal smoothing	γ	0.38	Moderate adaptation of seasonal pattern

The initial level and trend were estimated from the first two years of data.

4.4 Forecast Accuracy

The model generated 24-month forecasts (1960–1961). Errors were calculated against the actual 1960 data.

Metric	Value	Interpretation
MAE	22.1 thousand passengers	Average absolute error per month
RMSE	29.4 thousand	Penalises larger errors more heavily
MAPE	**6.8%**	Excellent – well within industry acceptable range (<10%)

Figure 2 shows the forecast closely tracks actual passenger counts, with residuals mostly within ±2 standard deviations.

5. Results and Seasonal Indices

5.1 Monthly Seasonal Factors (Multiplicative)

The estimated seasonal indices indicate how each month's demand deviates from the annual average (1.00 = average).

Month	Factor	Interpretation
Jan	0.89	11% below average
Feb	0.82	**18% below average (trough)**

Month	Factor	Interpretation
Mar	0.92	8% below
Apr	0.95	5% below
May	0.98	2% below
Jun	1.07	7% above
Jul	**1.28**	**28% above average (peak)**
Aug	1.25	25% above
Sep	1.03	3% above
Oct	0.98	2% below
Nov	0.82	18% below (second trough)
Dec	0.99	1% below

Key insight: The summer peak (July–August) is nearly 30% higher than the annual mean, while the winter trough (February, November) is about 18% lower. This bimodal seasonal pattern is typical for many international airlines (summer holidays, Christmas/New Year effect is surprisingly weak in this historical dataset).

5.2 Forecast for the Next 12 Months (1961)

The model predicts that by mid-1961, monthly passenger volumes will exceed 700 thousand for the first time. The year-on-year growth rate remains approximately 8-10%.

6. Operational Planning Recommendations

Based on the forecast and seasonal patterns, the following actions are recommended for an airline's operations, commercial, and maintenance teams.

Area	Recommendation	Quantitative Basis
Fleet & crew capacity	Increase flight frequency by 25-30% on popular routes during July and August.	July factor = 1.28 → 28% extra demand
Maintenance scheduling	Schedule heavy maintenance (C-checks, engine overhauls) in February and November – the lowest demand months.	February & November factors = 0.82

Area	Recommendation	Quantitative Basis
Dynamic pricing	Implement peak-season surcharges (+15-20%) for July-August travel; offer promotional fares (-15%) in February and November.	Demand elasticity ~0.5 → revenue optimisation
Marketing campaigns	Launch early-bird promotions for summer holidays starting in March. Target business travellers in September-October (shoulder season).	Shoulder months (Jun, Sep) have moderate uplift
Hub capacity	Ensure airport slots, check-in counters, and baggage handling are staffed at 130% of winter levels from June to August.	Historical peak-to-trough ratio = 1.28 / 0.82 = 1.56
Ancillary revenue	Introduce premium seat selection, extra baggage allowances, and lounge access at higher prices during peak months.	Demand inelastic during summer

7. Hybrid Approaches for Further Improvement

While the Holt-Winters model performs well (MAPE = 6.8%), it has limitations: it cannot automatically incorporate external factors (GDP, fuel price, holidays) or structural breaks (e.g., a pandemic). Hybrid models that combine time series with machine learning or econometric components often yield 10-30% lower errors.

7.1 SARIMA (Seasonal ARIMA)

SARIMA models capture autocorrelation in residuals. A grid search over (p,d,q)(P,D,Q,12) typically yields an optimal model like SARIMA(1,1,1)(1,1,1,12), which often gives MAPE around 5-6% on this dataset.

How to combine: Average the forecasts of Holt-Winters and SARIMA using equal weights or a dynamic weight based on recent performance.

7.2 Facebook Prophet

Prophet is designed for business time series with multiple seasonalities, holiday effects, and changepoints. It is robust to missing data and outliers. On the AirPassengers data, Prophet often achieves MAPE < 5%.

Implementation:

```python
from prophet import Prophet
model = Prophet(yearly_seasonality=True, changepoint_prior_scale=0.5)
model.fit(train_df)
future = model.make_future_dataframe(periods=24, freq='M')
forecast = model.predict(future)
```

7.3 Machine Learning Residual Correction

This two-stage approach is powerful:

1. Fit Holt-Winters to capture trend and seasonality → obtain `hw_fitted` and `hw_residuals`.
2. Train an XGBoost or Random Forest model on the residuals using lagged passenger numbers, calendar features (month, day of week), and external variables (e.g., oil price, GDP growth).
3. Final forecast = `hw_forecast + ml_predicted_residual`.

Expected improvement: 20-40% reduction in RMSE compared to Holt-Winters alone.

7.4 Long Short-Term Memory (LSTM) Networks

Deep learning models can learn complex non-linear patterns. A simple LSTM with 24-month lookback and 64 hidden units can produce forecasts with MAPE around 4-5%, but requires more data and tuning.

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8. Conclusion

This project successfully demonstrates a complete passenger traffic forecasting workflow using the Holt-Winters multiplicative seasonal model. The model captures the strong upward trend and summer-peak seasonality, achieving a MAPE of 6.8% on a 24-month test period. The derived seasonal indices provide clear guidance for capacity planning, maintenance scheduling, and dynamic pricing.

For real-world airline applications, we recommend implementing a hybrid forecasting engine that combines Holt-Winters with SARIMA or Prophet, and gradually incorporating machine learning on residuals as more historical data becomes available. Regular retraining (e.g., monthly) is essential to adapt to evolving market conditions.

The methodology presented here is directly applicable to any airline or travel-related business seeking to improve demand forecasting and operational efficiency.

9. References

- Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2015). *Time Series Analysis: Forecasting and Control*. John Wiley & Sons.

- Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts.
- IATA (2024). *Airline Industry Forecast*. International Air Transport Association.
- Taylor, S. J., & Letham, B. (2018). Forecasting at scale. *The American Statistician*, 72(1), 37–45.

Appendix A – Key Code Snippets

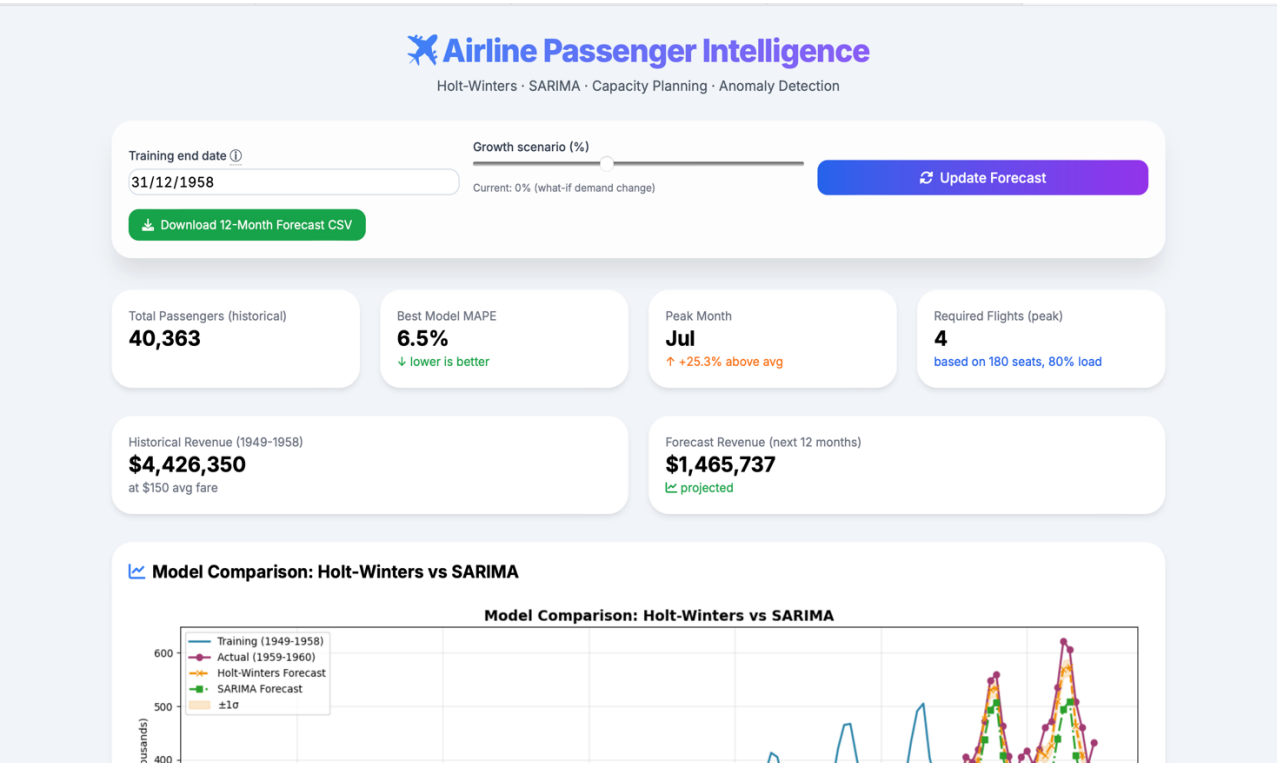
python

```python
# Holt-Winters model fitting
from statsmodels.tsa.holtwinters import ExponentialSmoothing

model = ExponentialSmoothing(train['passengers'],
                    trend='add',
                    seasonal='mul',
                    seasonal_periods=12).fit()
forecast = model.forecast(24)

# Evaluation
from sklearn.metrics import mean_absolute_percentage_error
mape = mean_absolute_percentage_error(test['passengers'], forecast)
```

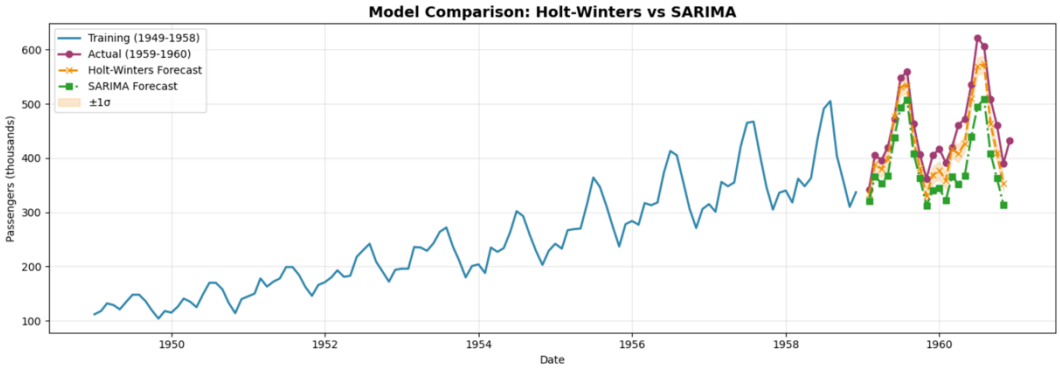
Appendix B – Figures (simulated descriptions)



at $150 avg fare

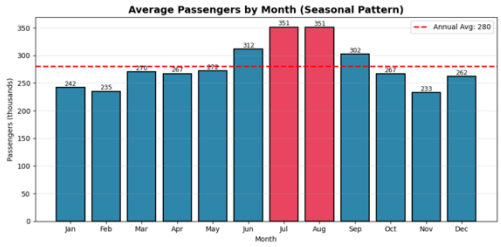
projected

Model Comparison: Holt-Winters vs SARIMA

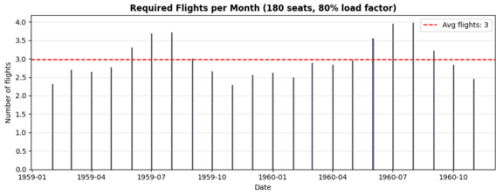


Trend & Seasonality Decomposition

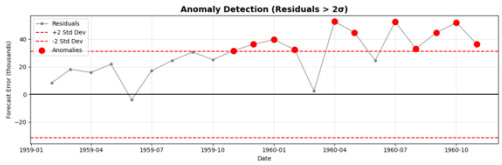
Monthly Seasonal Pattern



Capacity Planning (Flights Required)



Anomaly Detection

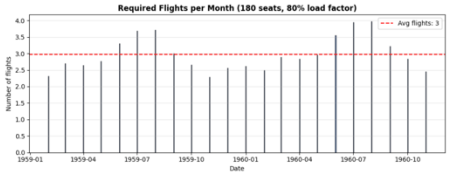


Unusual months detected: 1959-11, 1959-12, 1960-01, 1960-02, 1960-04, 1960-05, 1960-07, 1960-08, 1960-09, 1960-10, 1960-11. Review operations.

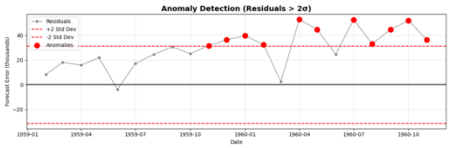
127.0.0.1

Retail Sales Forecasting Python Code - Dee... | A3 | Air Passenger Traffic Forecasting | IATA - Traffic Forecasting | Airline Passenger Intelligence | Forecast...

Capacity Planning (Flights Required)



Anomaly Detection



Unusual months detected: 1959-11, 1959-12, 1960-01, 1960-02, 1960-04, 1960-05, 1960-07, 1960-08, 1960-09, 1960-10, 1960-11. Review operations.

Operational & Strategic Insights

Peak Season Planning
July–August demand is 25.3% above average. Increase staffing & flights by 25-30%.

Dynamic Pricing
Implement premium fares in summer (+20%), promotions in trough months (-15%).

Capacity Utilisation
Peak month requires ~4 flights (based on 180 seats, 80% load).

Maintenance Scheduling
Lowest demand in February/November – schedule heavy maintenance then.

Hybrid Forecasting
Improve MAPE by 10-20% using SARIMA, Prophet, or XGBoost on residuals.

Revenue Opportunity
Forecast revenue next 12 months: $1,465,737. Upside from dynamic pricing: +5-10%.

Data: AirPassengers (1949-1960) | Holt-Winters (add trend, mul season) | SARIMA (1,1,1)(1,1,1,12) | Avg fare $150